

Environmental Management Services at Maitri, Antarctica

G. Veerabhadraiah and Anil Jain

Environment Division, Engineers India Ltd., New Delhi

Engineers India Limited (EIL) started its association with the Indian Antarctic Program by participating in the summer part of the 23rd Indian Antarctic Expedition (XXIII-IAE). EIL provided its services in reviewing Wastewater Management at Maitri, the permanent Indian Research Station in Antarctica. The work carried out by Environment Division of EIL for the study Wastewater Management at Maitri is covered here.

In order to carry out the study, a 10 point methodology has been adopted covering various aspects such as collecting front-end information, review of design data, obtaining operating and site specific information and their analysis to arrive at final conclusions and recommendations. Site work was initiated with collection of all available data/information with respect to existing wastewater management facilities. In order to get the design, operating and maintenance details of existing biodiscs treating the effluent from Maitri, EIL communicated with system suppliers i.e. Klargest Environmental Engineering, UK. A database has been created with all the information obtained and the same was made available to Expedition Leader. Also collected relevant data on wastewater generation and its handling associated with summer camp facilities. Historical operating data available from earlier expeditions has been compiled. Also collected site-specific data as well as feedback from leaders of past expeditions and inmates (wintering personnel) of Maitri to obtain the first hand information. At the end of the expedition, various interim measures have been suggested for up keeping of the wastewater management facilities pending the execution of modifications required, and the same were discussed and explained to Expedition Leader and Logistic Team.

After the expedition, the existing system has been reviewed with all the available information collected at site and the fresh data collected from NCAOR. Conclusions have been made highlighting various limitations of

the existing facilities along with the recommendation to overcome the same suggesting the action plan. While submitting the report, a presentation has also been made to the Technical Committee in this regard. Subsequently EIL has submitted proposal to implement the modifications required suggesting the methodology of execution in phased manner.

Engineers India Limited (EIL) having its headquarters in New Delhi and offices in India and abroad, is a premier consulting organization in South-east Asia which provides value-added services in varied fields of engineering using state-of-the-art technology. EIL is looking forward to actively contribute towards this mission of great importance to mankind and to the nation by offering its services for any engineering and technological requirements.

INTRODUCTION

Maitri - India's permanent Antarctic station was built in 1989 and is located in Schirmacher Oasis region in Princes Maud Land in East Antarctica. It serves for year round scientific exploration and research studies by various organizations. Every year, Indian expedition is conducted with scientists, engineers, and logistic personnel with half the team stay for only austral summer and the rest for the entire year. Maitri was designed for about 25 personnel with basic infrastructure and life support systems. Summer team stays in specially designed timber huts.

M/s National Centre for Antarctic & Ocean Research (NCAOR), Goa, under Department of Development (DOD), Government of India, is the nodal organization for the co-ordination and implementation of the Indian Antarctic Program including the up-keep of India's permanent (year round) station - Maitri at Antarctica. In 2003, M/s NCAOR invited Engineers India Limited (EIL), New Delhi to participate in Indian Antarctic Expedition to carryout structural assessment of Maitri and review of its wastewater management system. This report is about the work carried out by Environment Division of EIL for the review of wastewater management system.

FIELDS OF INVESTIGATION

The existing wastewater treatment units and associated facilities required detailed study for review and their improvement to meet the applicable effluent discharge standards inline with Madrid Protocol

requirements as well as to eliminate contamination of drinking water lake which is at lower elevation compared to the building and its effluent handling units.

Keeping in view the nature of the study concerned, following 10-Point methodology was adopted to carry out the study.

- Front-end information from DOD Reports
- Site visit for first hand information of existing facilities
- Review of Design information
- Collection of As-built data
- Collection of historical and recent operating data
- Visual observations & Feedback
- Collection of site specific data
- Evaluation & Analysis of the system
- Conclusions & Recommendations
- Nature of modifications required and suggested plan of action

SCIENTIFIC IMPORTANCE OF ANTARCTICA

Though the study carried out, per se, was out of logistic requirements for up keeping of the Maitri, it has got significant importance primarily to sustain the Indian Antarctic research besides utilizing the study results for all future Antarctic operations.

TITLES OF VARIOUS PROJECTS SO FAR UNDERTAKEN IN ANTARCTICA

Since it is for the first time EIL participated in the Indian Antarctic Program during 23rd Austral Summer, 'Review of Wastewater Management at Maitri' was only the project undertaken till date.

SCIENTIFIC AIMS OF OBJECTIVES IN ANTARCTICA

The project being an outcome of logistic requirement, the objectives of the work carried out are to come out with various limitations of existing wastewater management at Maitri in order to overcome the associated problems.

REGION OF INVESTIGATIONS IN ANTARCTICA

The project covers wastewater management aspects of Maitri including its associated summer facilities.

PROJECT WISE DETAILED REVIEW OF THE WORK CARRIED OUT

Various aspects of the work carried out under the study at site are as below:

- Set up of existing effluent handling facilities
- Specification of units
- Operating Data
- Visual Observations
- Feedback information
- Site Specific Data
- Regulatory Requirements

Details of the above are given in following sub sections.

SET UP OF EXISTING EFFLUENT HANDLING FACILITIES

Effluent Collection Arrangement

Raw effluent from kitchen wash basins is collected in 1½" CI pipe and then connected to 5" CI pipe which opens into a mesh trap to collect heavier solids. There from 3" CI pipe is laid connected to 3" PVC pipe before it leaves outside Kitchen to Klargesters B1 with electric heat tracing with PVC foam insulation.

Raw effluent from washbasin modules in toilet area is collected through 1" copper pipe which is connected to header of same size and material from bathroom modules, with PVC foam insulation. From there it runs without insulation while collecting effluent from urinal block.

A separate 1" copper pipe collects effluent from washing/laundry area, which combines with above line from toilet area combining into 1½" copper pipe in boiler room carrying all grey water to Klargesters B3 through 3" PVC pipe provided with electric heat tracing with PVC foam insulation.

TREATMENT DESCRIPTION

Klargester units are compact and incorporate primary settlement zone,

biozone and sludge zone in a single unit. These units work on aerobic bio treatment principle with rotating biological contactor (RBC) technique.

Primary Settlement Zone

Raw effluent is directly received in primary settlement zone through 4" piping inlet with baffle arrangement acting as stilling chamber. This helps in homogenizing the contents and settlement of heavier solids at the bottom while the effluent with lighter solids enters into the biozone through an entry pipe at its bottom.

Bio Zone

The biozone comprises of semi circular chamber and a motorized rotating disc incorporated with banks of polypropylene media sheets separated with transverse baffles so that the liquid follows a serpentine path through the zone. This disc is partially immersed in liquid in the chamber alternately coming in contact with effluent and ambient air as it rotates for intake of food and oxygen respectively. Naturally developed microorganisms adhere to the disc sheets to form a biologically active film feeding upon mainly the organic impurities and in the process multiply their population. Treated effluent is collected in outlet chamber where excess biomass is settled and clarified effluent is routed to "disposal through 4" piping outlet.

Sludge Zone

The settled solids from primary settlement zone and settled biomass from the outlet chamber collect in the base of the unit but in respective chambers. Settled sludge is required to be removed generally by a gully sucker or similar unit, as no drain points are provided.

There are some variations in the installed B1 & B3 and the un-installed BA units, which are detailed subsequently along with their dimensional details. Both the units are placed to the west of and beside Maitri's B-Block.

DISPOSAL ARRANGEMENT

Treated effluent from both the Klargesters i.e. B1 & B3 outlet is routed to Treated Effluent Pond adjacent to the klargesters, through dedicated 3" PVC pipes provided with electric heat tracing with PVC foam insulation.

The pond is open, earthen, unlined unit spread along side of the B-Block, with capacity catering to a period of 3-6 months of effluent routed to

Table 8.1—Specifications of Klargester Biodiscs

Item#	B1 (Kitchen)	B3(Grey Water) (uninstalled)	BA
Design Flow (m ³ /day)	1.2	5.0	1.2
Design Peak Flow (m ³ /hr)	0.2	0.93	0.2
Design BOD Load (kg/day)	0.36	1.5	0.36
Design Am.Nitrogen Load (kg/day)	0.048	0.144	0.048
Process	Aerobic Biodegradation		
Technique	Rotating Biological Contactor		
Make	Klargester Environment Ltd., UK		
Model	B1	B3	BA
Top Size (internal)	1.85 X 1.6	3.0 X 2.1	1.85 X 1.6
Top Size (external)	1.95 X 1.7	3.1 X 2.2	1.95 X 1.7
Total Height (internal)	1.9	1.8	1.9
Total Height (external)	2.1	1.8	2.1
Inlet Invert depth (free board)	0.75	0.28	0.75
No. of biozones	1	1	2
Availability Managed flow system	No	No	Yes
Length of biodisc	1.0	1.3	0.25 & 0.5
Dia. Of biodisc	0.5	1.4	0.9
Shape of media sheets	Square	Square	Circular
No. of sheets in first biozone	46	71	12
No. of sheets in second biozone	NA	NA	25
Size of each sheet	0.27	0.96	0.84
Recycle rate (m ³ /hr)	Nil	0.16	Nil
Size of Inlet/Outlet (inch)	4"	4"	4"
Material of Unit	GRP	GRP	GRP
Material of media	PP	PP	PP
PST Volume (m ³)	1.17	5.25	1.17
Biozone volume (m ³)	0.31	1.1	0.3
FST volume (m ³)	0.37	1.3	0.37
Motor Rating	60 Watts	60 Watts	60 Watts
Motor rpm	1440	1440	1440
Post gear rpm	5.7	5.7	2.5
Disc rpm	1.2	1.3	1.2
Bearing size (mm)	25	40	25
Empty (dry) weight (Tonnes)	0.35	1.3	0.35
Full weight (Tonnes)	2.4	9.5	2.4
Pressure Drop (fall across unit) (mm)	300	300	300
Desludging interval (months)	2-4	2-4	2-4

All dimensions are in meters unless specified.

* BA is similar to B1 except for the later having smaller media with two biozones in series and managed flow sysem

it. As and when the pond is filled, its contents are pumped out using portable pump about 250 m away from Maitri towards its south west side.

Treatment philosophy and its operation is depicted in Process Flow Diagram (PFD) (Drg No. 8517-00-17-41-0001 Rev.A) and Piping & Instrumentation Diagram (P&ID) (Drg No. 8517-00-17-41-0002 Rev.A) respectively.

SPECIFICATIONS OF THE UNITS

Specifications of the units are tabulated below which include the design data collected from KEE Ltd., and dimensional details collected at site, with physical measurements as well as available in-house data for the package units. Also, sketches have been prepared with relevant details collected at site before communicating with equipment supplier M/s Klargester, UK to get various information of what they had supplied.

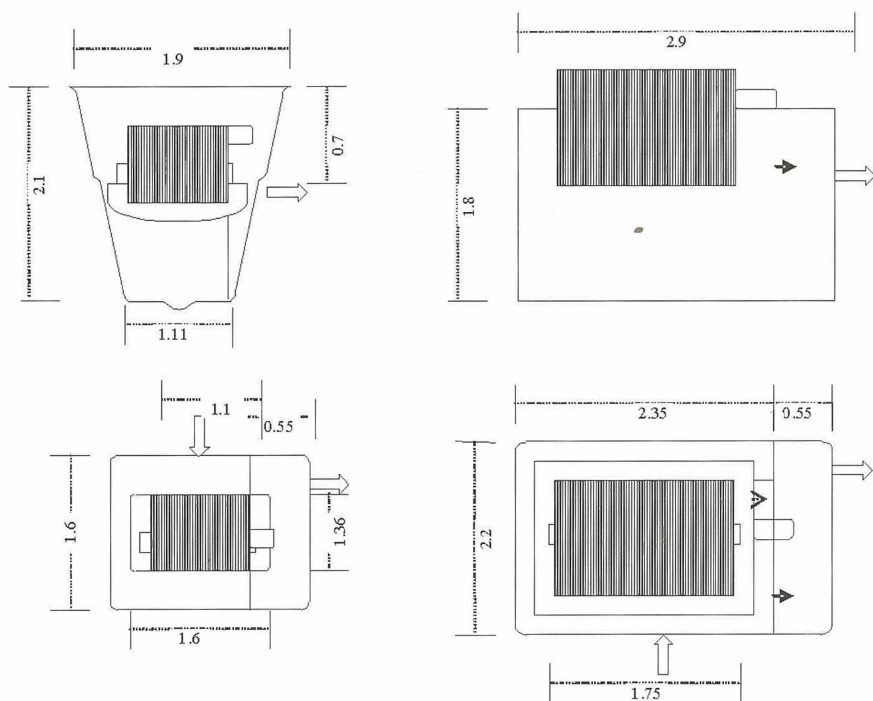


Fig. 8.1 : Indicative Sketches of Klargester Biodiscs

Table 8.2—Specifications of Effluent Pond

Item#	Effluent Pond
Volume (m3)	500
Top Dimensions (internal)	37 X 17 avg.
Top Dimensions (external)	40 X 20 avg.
Total Height	0.8
Inlet Invert depth (free board)	Entry from top
Dia. of biodisc	0.5
Shape	Parallelepiped Trapezoid

Table 8.3—Specifications of Portable Pumps

Item#	Effluent Pump	Dewatering Pump
Flow (m3/hr)	30	60
Head (m)	20	28
Speed (rpm)	2840	3600
Power (HP)	5	3
Type	Monoblock(Portable)	Centrifugal Self priming (Portable)
Make	Sharp Tools	Shriram Honda
Model	65 L 3.7	GK-200

OPERATING DATA

Operating data has been collected with respect to historical monitoring data available from various expedition reports and the data collected as part of the latest expeditions including the current one, and all the data has been tabulated, as below :

Table 8.4—Operating Data for B1 (Klargester for Kitchen Effluent)

Parameter	Historical						Re- cent 22 nd IAE	Current	
	13 th IAE		14 th IAE		19 th IAE			23 rd IAE	
	Inlet	Out- let	Inlet	Out- let	Inlet	Outlet		Inlet	Outlet
pH	8	5.5	4.4	4.3	5.1	5.1		5	4.5
TSS	646	124	148	112	97	20		1188	131
BOD	490	310	860	444	1363	454		620	320
COD	1510	920	2586	1332	2664	828		728	548
Sul- phates	18.5	10.6							
Nitrates	8.3	6.4	9.3	10.6	BDL	BDL			
Nitrites			0.008	0.01					
Amm. Nitro- gen			2.5	2.5	4.8	4.3			
Phos- phates	0.52	0.73			1.8	2.8			
Oil & Grease			1.1	ND	20.6	2.8			
Tempe- rature					8	6		22	21

VISUAL OBSERVATIONS

During site visit of effluent treatment facilities, one of the front-end activities carried out at Maitri, following observations have been made with respect to the system built, operation and maintenance.

System Built

- Pipes carrying grey water i.e. raw effluent from toilet modules and washing machines are of smaller than 1½".
- No insulation is available for major length of the pipes within the building.
- Support arrangement for the pipes running outside the building to B1 & B3 and their outlet lines reaching Pond comprises of drums, stones and is in haphazard manner.
- Non-availability of drain/sludge outlets for sludge withdrawal of the units.

Table - 8.5 Operating Data for B3 (Klargester for Grey Water)

Parameter	Historical						Recent		Current	
	13 th IAE		14 th IAE		19 th IAE		22 nd IAE		23 rd IAE	
	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet	Inlet	Outlet
pH	8.1	6.5	7.9	7.9	7.2	6.9			7	7.3
TSS	68	30	53	53	41	26			291	51
BOD	540	400	620	430	158	114			120	100
COD	1650	1100	1832	1296	522	306			280	208
Sulphates	55	97								
Nitrates	24.3	16.8	23	22.5	3.18	2.45				
Nitrites			0.01	0.03						
Ammonia-Nitrogen			100	100	122.5	47.5				
Phosphates	4.2	5.2			0.3	1.5				
Oil & Grease			12.8	5.2	5.8	4.2				
Temperature					8	6			21	21

- Resting of B1 & B3 housing shelters on ground level invites snow accumulation around them as well as on piping during snow-drift.
- Both B1 & B3 housing has no adequate room inside around the units to facilitate easy access and maintenance of the units.
- Proper approach is not available to access different portions of the units.

Operation

- Major contamination load is from liquid and powder detergents used in Kitchen and Laundry facilities.
- Wide fluctuations in generation of effluents.
- Build-up of Oil/Fat layer in B1.
- Milky-white to Grey color biomass development on klargesters bio discs media.
- Excessive sloughing of biomass, especially in B1.

- Temperature control through 500 W bulb located inside each unit.
- Stinking septic smell from both units, more pungent from B1.
- Heavy sludge buildup/accumulation in both the units.
- Carry-over of white/grey biomass from final settlement chamber of the units.
- Sludge bulking (floating of sludge) in final settlement chamber of B1.
- Allowing effluent to be accumulated in the Pond and pumping of the same out to discharge location only twice during the summer period.
- Unacceptable smell and colour of the Pond contents.
- Freezing of pond effluent due to open and large surface area of the Pond.
- Running of seepage channel from Pond and later combining with snow melt water from upstream, to Priyadarshini Lake.
- Non-existence of any monitoring facilities for unit loads and operation.

Maintenance

- Reports of chokage of effluent lines from bathroom modules have been noted requiring frequent air flushing.
- PVC pipes running in outdoors were found sagged with improper connections and significant leakages have been observed from both influent and effluent lines of B1 & B3.
- Manual sludge withdrawal using bucket arrangement, which is failed due to non-accessibility of the same into primary settling chambers of the units. Subsequent attempts by pumping also unsuccessful since the pumps available are mainly meant for clean liquid service. Bins collected with sewage sludge are left around the units in haphazard manner.
- Housing panels have been damaged at many places.

FEEDBACK INFORMATION

Following information has been gathered from earlier reports and from past expedition leaders and wintered members.

- Bathroom outlet pipes get blocked often and their cleaning is a problem due to space limitation around modules.
- Inlet and outlet pipes of klargetsers get damaged due to blizzards and had to be replaced/repared. The outlet pipes from klargesters get completely buried under snow during blizzards and needs clearing. Pipe leakages often take place and dirty water gets collected below Maitri due to ground slope. Maintaining the pipes has become a critical activity.
- As only temporary supports exist for the pipes, the trace heat coils fail, when the pipes buckle and give way at joints. A constant and vigilant watch needs to be kept on the trace heating connected to the Klargesters pipelines. Short-circuiting in the trace heating is not uncommon.
- Klargesters shelters fill with snow and water up to 4 ft height and needs clearing in winter periods. They have been replaced earlier as the old ones had been damaged and extensively leaking. Maintaining suitable conditions for klargesters is felt a burden.
- Frozen soakpits causes hindrance to the outflow of klargesters in winter period and results in unhygienic environment beneath and around the Maitri. Pond's ice is dug up and cleared to accommodate fresh load from Klargesters.
- Delay in pumping out pond water results in seepage as perma frost with 0.06 -1.0 m depth. Problem of waste disposal prevails almost entire year. With on-set of summer, the melting ice from surroundings also feeds water into this pond at a very large rate and along with pond water reaches Lake. It is a challenge in itself to maintain the level of the pond.

SITE SPECIFIC DATA

Locational Details Using GPS

Various other relevant data has been collected as given below :

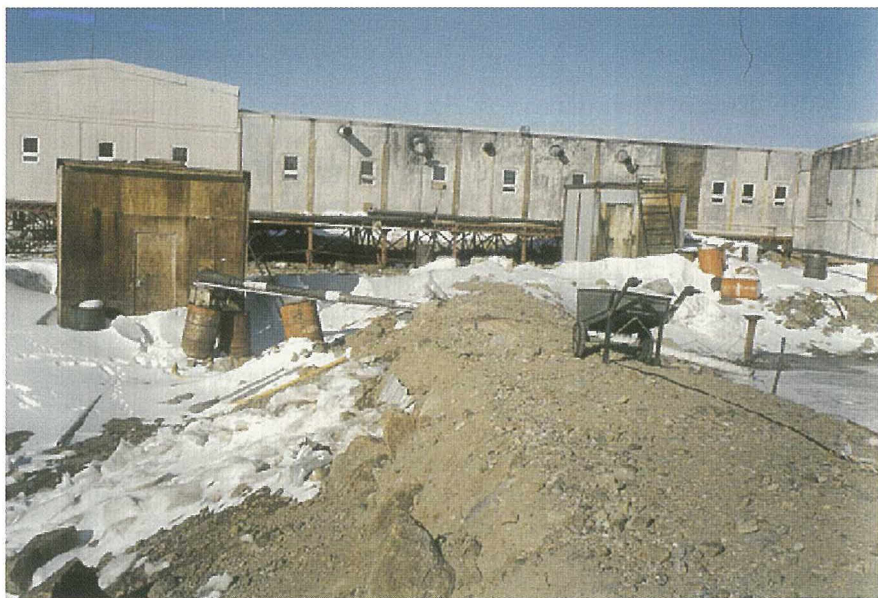
Table 8.6—GPS Data for Environmental & Associated Systems

Item	1	2	3	4	5	6	7
Latitude 70°+ S (' °)	4600.4	4603.6	4603.5	4552.5	4605.2	4555.1	4553.0
Longitude 11°+ E (' °)	4349.1	4331.3	4311.1	4324.8	4347.4	4346.4	4343.9
Altitude (m)	126	128	119	116	127	116	116
Distance from '1' (m)	0	210	400	350	150	170	240

1. WW Pond
2. Present Disposal Point
3. Present Disposal Lake
4. Present Zub Lake Meeting Point
5. Fuel Tank
6. DW Line Shore Point
7. DW Line Pumping Point

LAYOUT DETAILS

Details of layout of environmental treatment facilities with respect to Maitri have been collected and depicted in **Layout Diagram (Drg No. 8517-00-17-41-0003 Rev.A)** attached herewith.



Wastewater treatment system adjacent to Maitri's B-Block

Regulatory Requirements

Antarctica Treaty Requirements

As it is known, all Antarctic operations are carried out by member nations under the regime of 'Antarctic Treaty', formed in December 1959. All member nations are signatories to the 'Protocol on Environmental Protection', which came into existence in 1991 and finally into force in 1998.

In most cases, the protocol requirements associated with environmental management and its monitoring arise from the implementation of the same by national legislation. However there are legal requirements of protocol itself, to be adhered to by all member nations. Prime requirements of the protocol with respect to effluent disposal are:

- The principles contained in Article-3 which include the need to limit adverse impacts on the Antarctic environment including on water quality; and
- The specific waste management aspects contained in Annexure-III including the need to minimize the impacts of waste disposal and the requirement to discharge sewage and domestic liquid wastes wherever practicable where conditions exist for initial dilution and rapid dispersal.

National Legislation

Ministry of Environment & Forests (MoEF), Government of India issued "General Standards for discharge of environmental pollutants" in 1993 under the Notification on Environmental Standards under Environment (Protection) Act, 1986, preventing the discharges or emissions of environmental pollutants in excess of the prescribed standards. As per the Notification, effluent standards applicable for their discharge to 'Inland Surface Water' is given under Schedule-VI, Part-A : Effluents, are given in Table (8.7).

MONITORING REQUIREMENTS

"Environmental Monitoring is a fundamental element of basic research, environmental management, and conservation." as succinctly stated by SCAR/COMNAP, delineates following requirements with respect to wastewater management.

Table 8.7—Effluent Discharge Limits

Parameter	Allowable limit (mg/l)
Color & Odor	*
Suspended Solids	100
pH	5.5 to 9.0
Oil & Grease,	10
Total residual Chlorine	1.0
Ammonical Nitrogen (As N)	50
Total Kjeldahl Nitrogen (as N)	100
Free Ammonia (as NH ₃)	5.0
Biochemical Oxygen Demand (BOD ₅)	30
Chemical Oxygen Demand (COD)	250
Dissolved Phosphates (as P)	5.0
Sulphide (as S)	2.0
Phenolic Compounds (as C ₆ H ₆ OH)	1.0
Nitrate Nitrogen	10

* All efforts shall be made to remove color and unpleasant odor as far as practicable

- Each station should document the mass emissions of suspended solids, BOD, phosphorus and nitrogen in wastewater.
- For stations with fewer than 20 people, this can be estimated with sufficient accuracy, on the basis of models based on daily emissions per person.
- For stations with 20-200 people the modeled estimates should be augmented by actual measurements of slow rates each year and analysis of composite samples.
- For larger stations monthly measurements may be needed. If the actual measurements confirm the modeled estimates, they need not be continued.
- Also listed principal physical and chemical indicators of impact due to human activities: flow rate, TSS, BOD, pH, fecal coliforms, nutrients, and TKN.

System Evaluation & Analysis

Considering various data and information collected, a detailed analysis has been carried out for entire wastewater management system at Maitri, as below :

Table 8.8—Actual Outlet Quality

Parameter	Expedition wise										Desired Quality
	B1					B3					
	13 th	14 th	19 th	22 nd	23 rd	13 th	14 th	19 th	22 nd	23 rd	
PH	5.5	4.3	5.1	NA	4.5	6.5	7.9	6.9	NA	7.3	5.5-9.0
TSS	124	112	20	NA	131	30	53	26	NA	51	20
BOD	310	444	454	NA	320	400	430	114	NA	100	30
COD	920	1332	828	NA	548	1100	1296	306	NA	208	250

In order to study the system as to why it is not giving the desired results, various symptomatic and diagnostic aspects have also been scrutinized for studying the system deficiencies, while carrying out the analysis, as elaborated below:

Using the available operating data for both units, following graphs have been prepared for main parameters like pH, TSS, BOD and COD for each unit indicating their performance i.e. raw effluent characteristics vs. desired outlet limits:

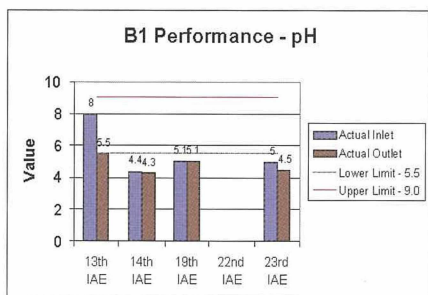


Fig. 8.2

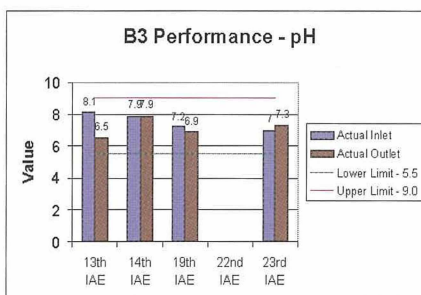


Fig. 8.3

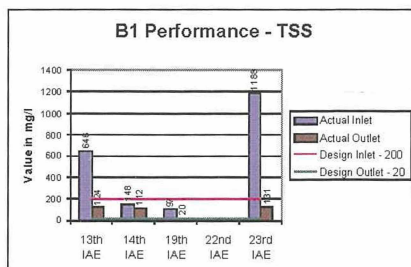


Fig. 8.4

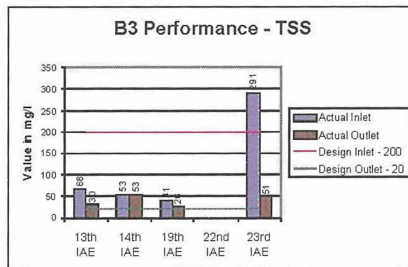


Fig. 8.5

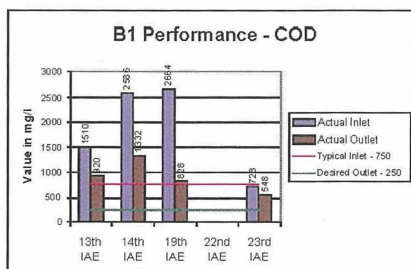


Fig. 8.6

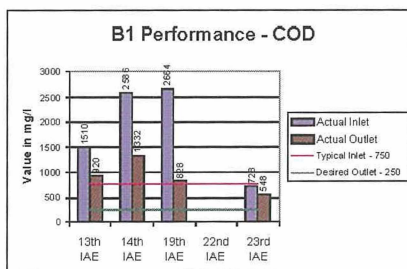


Fig. 8.7

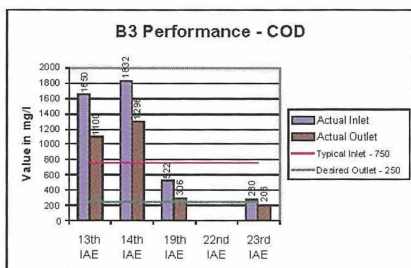


Fig. 8.8

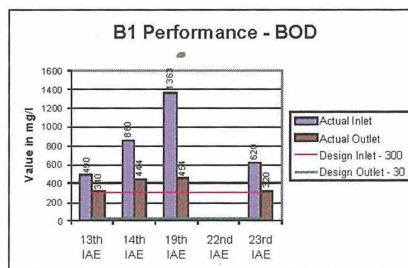


Fig. 8.9

CHECK ON UNITS LOADINGS

Table 8.9—Hydraulic & Pollutant Loads

Parameter	B1			B3		
	Operating		Design	Operating		Design
	Avg.	Max		Avg.	Max	
Hydraulic Load (m ³ /day)	0.9	0.9	1.2	2.5	3.0	5.0
BOD Load (kg/day)	0.75	1.23	0.36	0.9	1.86	1.5
TSS Load (kg/day)	0.47	1.07	0.24	0.28	0.87	1.0
Am.Nitrogen Load (kg/day)	0.003	0.004	0.05	0.28	0.37	0.15

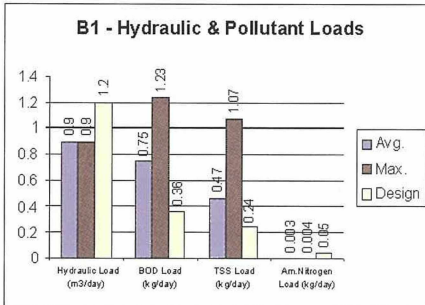


Fig. 8.10

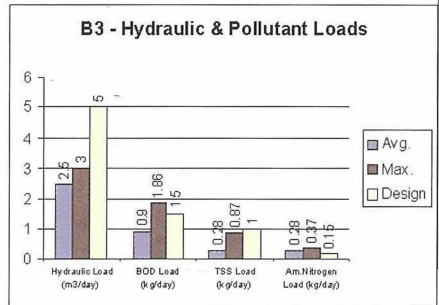


Fig. 8.11

CHECK FOR SHOCK LOADINGS

Table 8.10—Shock/Toxic Loads

Parameter	B1			B3		
	Operating		Design	Operating		Design
	Avg.	Max		Avg.	Max	
Peak Flow (m ³ /hr)	0.16		0.2	0.74		0.93
COD/BOD ratio	2.2	2	2*	3	3	*
% of Detergent Load**	30-50		25%	80-90		25%

* Typical for sewage. ** Based on observation of usage of the effluent sources

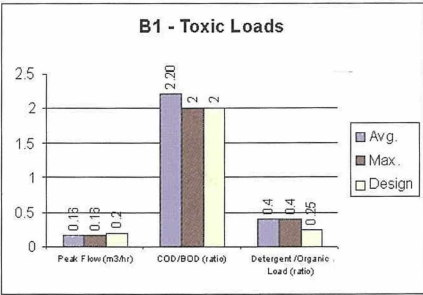


Fig. 8.12

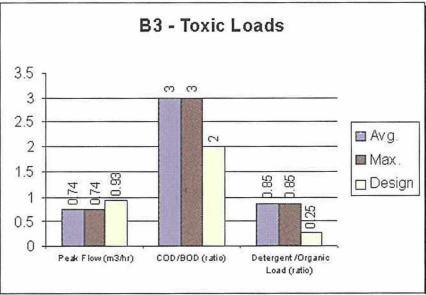


Fig. 8.13

CHECK FOR SLUDGE BUILDUP

Table 11.4—Sludge Buildup

Parameter	B1			B3		
	Operating		Design	Operating		Design
	Avg.	Max		Avg.	Max	
Peak Flow (m ³ /hr)	0.16		0.2	0.74		0.93
COD/BOD ratio	2.2	2	2*	3	3	*
% of Detergent Load**	30-50		25%	80-90		25%

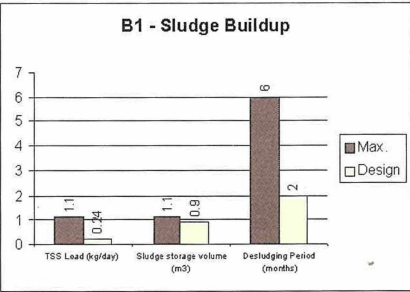


Fig. 8.14

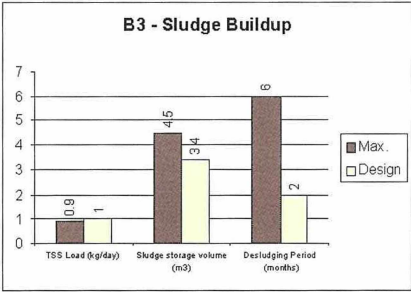
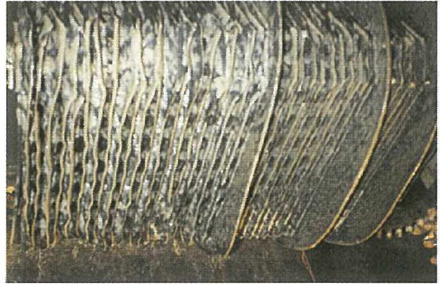


Fig. 8.15

CHECK ON PROCESS OPERATING CONDITIONS

Maintaining optimum

Temperature : Any aerobic biological system needs proper temperature maintaining between 25-34°C for optimum biomass cell growth and its proliferation. However below 13°C, their activity is expected to be severely



White/grey biomass on RBC's plastic media

diminished while the system shall never be operated below 5°C. It is found that maintain the right temperature within the units housing is an impossible/arduous task, despite usage of high wattage bulb in each of them, considering the material for housing and its status and the average daily ambient temperature of -20 to 4°C.

Check on pH of the biozone

contents: Aerobic biological treatment units need maintaining the pH of the effluent in between 6.5-8.5 and it is always desired to receive the effluent in alkaline range to take care of the pH reducing reactions taken place within the system. Though the pH of the raw effluent entering



Sludge bulking in Final settling tank of B1

the units are expected to be normally in line with the requirements, monitored data indicate significantly lower pH effluent in the inlet chamber of the units.

Availability of Dissolved Oxygen (DO): Aerobic systems need maintaining the minimum DO of 2 mg/l in the contents. In case of limiting of natural ventilation and/or higher organic loads, it is customary to provide supplement air supply facilities, which may be required for the subject conditions.

Conceptual Limitations : Separate handling of kitchen effluent and grey

water: The concept of segregation of kitchen from bath/laundry effluents is understandable but treating them in separate units is not called for while they are given similar treatment.

Lack of sludge withdrawal facilities : It is a basic limitation not to have the drain/sludge outlets for gravity sludge withdrawal or suitable sludge pumping arrangement.



Floating of oil & fat in primary settling tank

Unlined Pond : Since actual disposal point is far from the Treated Effluent Pond, It is essential for the same to have a protected walls to prevent percolation of the effluent to soil/water below it, infiltration of the water from below soil, overflow of its contents and entry of surface water into itself and the existing pond is not able to meet these criteria.



Effluent Pond after the contents are pumped out



Freezing of effluent in Effluent Pond

Improper Disposal Arrangement : The existing arrangement comprises of make shift pump to draw water from the Pond and conveying it to disposal point at 250m away from it, through a fire hose, as a typical stop gap arrangement.



Typical feeding arrangement to the Effluent Pond



Proximity of Effluent Pond to Priyadarshini Lake

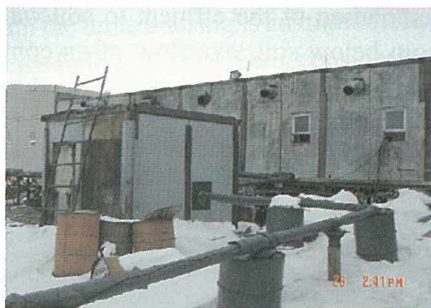
AS BUILT LIMITATIONS

Ineffective Housing Shelters: The existing dedicated housing shelters were not from the unit suppliers nor they were meant for the units and were executed as stopgap arrangement from discarded panels of Maitri. Due to installation of the units on ground level, the shelters are subjected to the wrath of harsh weather over the time and the panels have been damaged. Space available inside the shelters is hardly adequate even for regular operation of the units, leave alone the maintenance requirements.



B1 Housing Shelter

Improper Piping Support : Existing arrangement comprising discarded drums and small boulders is a haphazard one and is sensitive to maintenance for prevailing weather conditions and also pose a threat to hygiene of the surroundings.



Typical external piping arrangement

EXTERNAL LIMITATIONS

The following aspects, though do not affect the system performance directly, but play a sensitive role in meeting the objectives of wastewater management system.

Lack of Awareness : In order to maintain any system, it is essential to impart awareness among not only the operators but also the people for whom it is catering to. It is observed that no adequate information is made available to the logistic team vis-à-vis the training imparted to them with respect to other life supporting units.

Lack of Monitoring : Availability of adequate effluent monitoring facilities play a supporting role in maintaining whole the system in order, and further helps in meeting the regulatory requirements as well. Newly set up Environment Impact Assessment (EIA) Laboratory will go a long way in meeting the desired objectives.

PROJECT WISE NATIONAL STATUS

While collecting the historical data/information, it has been noted that the existing problems have been occurring since long (many years) and various national research and academic institutes had carried out studies in this respect. However, it was noted that the studies concluded only with reporting the then monitoring (system inlet & outlet quality) data and emphasizing on the requirement to modify the same.

But, in the study carried out by EIL, various aspects have been covered thoroughly in unprecedented manner viz. Establishing the design and as-built data by communicating with the system supplier, scrutinizing all available expedition reports as well as handing over reports for collecting historical data/information since Maitri's inception itself, detailed visual observations, obtaining feed back data, collecting site specific information and making it available in Expedition Leader's room. Based on the data, exhaustive evaluation and analysis of the system has been carried out to come out with Conclusions, recommendations and plan of action to implement the revamp of wastewater management facilities required at Maitri.

PROJECT WISE INTERNATIONAL STATUS

During the course of the study, information has been gathered with respect to systems operating at various Antarctic stations and they vary in their type depending on the location and number of personnel stationed. However it is to be noted that competitive units are presently also available in India. Utilizing the vast experience of EIL in the field, it has been proposed to explore indigenous package sewage treatment units to arrive at a suitable system for Maitri based on their track record.

PROJECT WISE MAJOR FINDINGS/OUTCOME

Following conclusions were drawn from the study.

- i. The effluent being discharged is highly polluted and far from the desired quality. Review and analysis of existing wastewater management system indicate that various factors are responsible to it.
- ii. One of the primary reasons for not meeting the desired effluent quality is high pollutant concentration in raw effluents exceeding their respective design limits.

- iii. In addition to the pollutant over loadings, toxic substances like high loads of detergents and associated high COD/BOD ratios cause inhibitory effect for biomass reproduction.
- iv. Segregated handling of kitchen effluent and grey water right up to the treatment is not called for. Combining both the streams would have dampened the excessive organic/solids loadings.
- v. An important factor causing the system inefficiency is excessive sludge buildups in the units, due to lack of sludge withdrawal facility inbuilt in the units or external arrangement to take care of the same.
- vi. The need to maintain optimum temperature seems to have been disregarded during installation of the system by locating the on-ground and housing the units with discarded panels.
- vii. Because of excessive organic pollutant loadings, system ability to provide adequate oxygen supply through natural aeration becomes limiting and results in development of septic conditions.
- viii. Unlined and open Effluent Pond, causes freezing of the top content and seepage/infiltration of the polluted water to the Priyadarshini (Zub) lake.
- ix. Type of Pond and make shift arrangement for disposing the pond effluent using a portable pump makes it workable only in summer periods and also proves to be ineffective to empty out the pond.
- x. Non-availability of enough room around the klargestar units within their housing block makes it difficult for their regular maintenance.
- xi. The condition of outdoor piping feeding to the klargestar units as well as that to the Pond looks appalling and definitely not suitable to the weather conditions.
- xii. Awareness about the existing system and its operation and maintenance requirements seem to be relatively lacking with the logistic team.
- xiii. Direct discharge of untreated effluent from various summer camp modules to priyadarshini lake is not an environmentally acceptable practice and is causing the surroundings unhygienic.

In light of the above, it is necessary to upgrade the wastewater facilities at Maitri.

OVER ALL CONTRIBUTION AND SIGNIFICANT ACHIEVEMENT

EIL carried out detailed and elaborated study of existing effluent handling facilities with following highlights.

- Collecting historical data/information
- Establishing the Design / As-built data
- Site-specific data using GPS
- Obtaining valuable Feed back from Maitrians
- Database of the information collected
- Detailed Analysis of existing system
- Arriving at revamp requirements
- Recommending Plan of Action and methodology of the revamp

Most Significant Achievement

Detailed study identifying all the limitations of the existing system with required modifications and suggesting plan of action with methodology of implementation.

RELAENAVE OF THE ACHIEVEMENT IN NATIONAL PERSPECTIVE AND FUTURE GROWTH OF THE COUNTRY

Implementing the modifications suggested will help in maintaining the proper environment and hygiene in and around Maitri and meeting the Environmental Protocol requirements of Antarctic Treaty. This will also make Maitri to have a state of the art wastewater management system.